The Impossibility of Faithful Explanation Under Underdetermination

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Abstract

When a system is underspecified—when multiple configurations produce the same observable output—any explanation faces an inherent tradeoff. We prove that no explanation can simultaneously be faithful (consistent with the system's native explanation), stable (invariant across observationally equivalent configurations), and decisive (committing to every distinction the native explanation makes). This impossibility holds whenever the Rashomon property obtains: equivalent configurations with incompatible explanations. We prove the Rashomon property is both necessary and sufficient—within our framework, the exact boundary between possibility and impossibility. The theorem is instantiated across nine domains: feature attribution, attention maps, counterfactual explanations, concept probes, causal discovery, model selection, saliency maps, token citation explanations, and mechanistic interpretability. For causal discovery, the Rashomon property is derived from first principles (Markov equivalence). The resolution—G-invariant aggregation—is proved Pareto-optimal among stable methods. The result is mechanically verified in the Lean 4 proof assistant (349 theorems, 0 unproved goals).

Significance. Explainability is central to trustworthy AI, scientific inference, and regulatory compliance. We prove a universal impossibility: no method of explaining an underspecified system can simultaneously be faithful to the system, stable across equivalent configurations, and decisive in its commitments. This formalizes a core consequence of the Quine–Duhem underdetermination thesis as a mathematical theorem, to our knowledge for the first time. The impossibility applies to nine major explanation paradigms in machine learning—from SHAP values to neural network circuit analysis—and extends to scientific explanation and medical diagnosis. The constructive resolution (aggregate over equivalent configurations) provides a Pareto-optimal tradeoff among stable explanation methods. The entire proof is mechanically verified, eliminating logical gaps in the formal proofs.

Keywords: impossibility theorem, explainability, underdetermination, Rashomon effect, formal verification

1 Introduction

When multiple configurations of a system produce identical observable output, any attempt to explain the system's behavior faces a fundamental constraint. This paper proves that constraint is an impossibility: no explanation can simultaneously be faithful, stable, and decisive.

This structure arises across science. In philosophy of science, Quine Quine [1951] and Duhem Duhem [1954] argued that empirical evidence underdetermines theory choice—no finite body of observations can uniquely select a scientific theory from among empirically adequate alternatives. In neuroscience, Marder and Goaillard Marder and Goaillard [2006] documented that nervous systems with identical functional output are built from qualitatively different circuit configurations, a phenomenon called *neural degeneracy* Edelman and Gally [2001]. In statistics and machine learning, Breiman's Rashomon effect Breiman [2001] and its modern elaboration by Rudin et al. Rudin et al. [2024] describe the multiplicity of near-optimal models with divergent internal structure. In economics, Manski Manski [2003] developed partial identification theory precisely because observational data generically fail to pin down a unique model. In database theory, Bancilhon and Spyratos Bancilhon and Spyratos [1981] proved that view updates are inherently ambiguous when the view is not injective.

We prove the impossibility, characterize its exact boundary (the Rashomon property—necessary and sufficient), demonstrate it across nine domains with the most detailed treatment in machine learning, and provide a constructive resolution. Informally:

> *If two configurations of a system produce the same observable output but honestly report incompatible explanations, then no explanation method can simultaneously be faithful to the system, stable across equivalent configurations, and decisive in its commitments.*

The proof is four steps—decisiveness, stability, faithfulness, contradiction—and is mechanically verified in the Lean 4

proof assistant de Moura and Ullrich [2021] (74 files, 349 theorems, 72 axioms, 0 `sorry`). The core impossibility requires zero axioms beyond the Rashomon property itself.

Our contributions are: (1) To our knowledge, this is a universal impossibility theorem unifying nine domain-specific results under a single proof; (2) a proof that the Rashomon property is both necessary and sufficient—the exact boundary between possibility and impossibility; (3) nine instances spanning feature attribution, attention maps, counterfactual explanations, concept probes, causal discovery (with Rashomon *derived* from Markov equivalence), model selection, saliency maps, token citation explanations, and mechanistic interpretability; (4) a constructive resolution via G-invariant aggregation proved uniquely optimal; (5) empirical validation across seven instances; and (6) complete mechanical verification. The result connects to Hunt–Stein invariant decision theory Lehmann and Romano [2005] and, to our knowledge, provides the first mathematical formalization of the Quine–Duhem underdetermination thesis.

2 Framework

2.1 Explanation Systems

We introduce a minimal abstract framework in which the impossibility can be stated and proved. All nine instances in Section 3 are special cases.

Definition 1 (Explanation System)**.** *An* explanation system *is a tuple* $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{observe}, \mathsf{explain}, \perp)$ *where:*

- $\mathbf{\Theta}$ *is a* configuration space*—the set of possible configurations of the system (e.g., model parameters, experimental setups, circuit wirings);*
- $\mathcal{H}$ *is an* explanation space*—the set of possible explanations;*
- Y *is an* observable space*—the set of measurable outputs;*
- $\mathsf{observe} : \mathbf{\Theta} \to Y$ *assigns to each configuration its observable behavior;*
- $\mathsf{explain} : \mathbf{\Theta} \to \mathcal{H}$ *assigns to each configuration the explanation it "honestly reports"; and*
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ *is an* incompatibility relation *satisfying* irreflexivity*:* $\neg \perp (h, h)$ *for every* $h \in \mathcal{H}$*.*

An *explanation map* $E : \mathbf{\Theta} \to \mathcal{H}$ is a procedure that a practitioner *chooses*. The native explanation $\mathsf{explain}(\theta)$ records what the configuration actually implies; $E(\theta)$ is what the chosen method reports.

2.2 Three Desiderata

Definition 2 (Faithful, Stable, Decisive)**.** *An explanation map* E *is:*

- ***Faithful*** *if* $E(\theta)$ *never contradicts the native explanation:* $\forall\, \theta,\ \neg \perp (E(\theta), \mathsf{explain}(\theta))$*.*
- ***Stable*** *if equivalent configurations receive the same explanation:* $\forall\, \theta_1, \theta_2,\ \mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \Rightarrow E(\theta_1) = E(\theta_2)$*.*
- ***Decisive*** *if* $E(\theta)$ *inherits every incompatibility of the native explanation:* $\forall\, \theta, h,\ \perp (\mathsf{explain}(\theta), h) \Rightarrow \perp (E(\theta), h)$*.*

Faithfulness says the method does not contradict the system. Stability says the method is reproducible—it does not depend on which equivalent configuration happens to be examined. Decisiveness says the method is at least as committal as the system itself: whatever the native explanation rules out, the chosen explanation also rules out.

2.3 The Rashomon Property

Definition 3 (Rashomon Property)**.** *An explanation system has the* Rashomon property *if there exist* $\theta_1, \theta_2 \in \mathbf{\Theta}$ *with*

$$\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2) \quad \textit{and} \quad \perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2)).$$

Two configurations produce the same observable output yet honestly report incompatible explanations. This is the formal statement of underdetermination at the explanation level.

2.4 Main Theorem

Theorem 4 (Universal Explanation Impossibility)**.** *If an explanation system has the Rashomon property, then no explanation map* E *can simultaneously be faithful, stable, and decisive.*

Proof. Let θ_1, θ_2 witness the Rashomon property. Suppose E is faithful, stable, and decisive.

1. **Decisiveness** at θ_1: $\perp (\mathsf{explain}(\theta_1), \mathsf{explain}(\theta_2))$ implies $\perp (E(\theta_1), \mathsf{explain}(\theta_2))$.
2. **Stability**: $\mathsf{observe}(\theta_1) = \mathsf{observe}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$, so $\perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
3. **Faithfulness** at θ_2: $\neg \perp (E(\theta_2), \mathsf{explain}(\theta_2))$.
4. Steps 2 and 3 contradict. □

Proposition 5 (Tightness)**.** *Each pair of properties is simultaneously achievable: (i) faithful + decisive (set* $E = \mathsf{explain}$*; stability fails); (ii) faithful + stable (set* E *to a neutral constant; decisiveness fails); (iii) stable + decisive (set* E *to a maximally committal constant; faithfulness fails).*

Proposition 6 (Necessity)**.** *An explanation system admits a faithful, stable, and decisive explanation map if and only if it does* not *have the Rashomon property.*

The Rashomon property is therefore the *exact boundary* between possibility and impossibility. The proofs of Propositions 5–6 are in *SI Appendix*.

3 Instances

We instantiate Theorem 4 in four domains chosen for cross-disciplinary breadth. Five additional instances (attention maps, counterfactual explanations, concept probes, model selection, language model explanations) are detailed in *SI Appendix*.

3.1 Feature Attribution (Machine Learning)

The 6-tuple: $\mathbf{\Theta}$ = trained models (GBDT, Lasso, neural networks); Y = input–output function; $\mathcal{H}$ = attribution vectors $(\varphi_1, \ldots, \varphi_P)$; $\perp$ = opposite ranking of a collinear pair. Collinearity induces a Rashomon set of near-optimal models differing only in which correlated feature acts as primary predictor D'Amour et al. [2022], Fisher et al. [2019]. The impossibility follows from Theorem 4, subsuming Bilodeau et al. [2024] as a special case.

Quantitatively: for GBDTs the attribution ratio diverges as $1/(1-\rho^2)$; for Lasso one feature is zeroed entirely (infinite ratio). Empirically, SHAP rankings flip at a 33% rate across equivalent models.

3.2 Causal Discovery

The 6-tuple: $\mathbf{\Theta}$ = DAGs over vertex set V; Y = conditional independence (CI) structures; $\mathsf{observe}$ = d-separation Markov condition; $\mathcal{H}$ = fully oriented DAGs; $\perp$ = opposite orientation of some edge. Two DAGs are *Markov equivalent* iff they encode the same CI structure Verma and Pearl [1991].

Proposition 7 (Causal Rashomon Property—Derived)**.** *Any CPDAG with an undirected edge $\{X, Y\}$ witnesses the Rashomon property: orient $X \to Y$ for h and $Y \to X$ for h'; both are Markov-consistent yet $\perp (h, h')$.*

By Theorem 4, no orientation rule can simultaneously be faithful, stable, and decisive. The resolution is to report CPDAGs Chickering [2002]—trading decisiveness for faithfulness and stability.

3.3 Mechanistic Interpretability

The 6-tuple: $\mathbf{\Theta}$ = weight configurations; Y = input–output function; $\mathcal{H}$ = circuit decompositions; $\perp$ = contradictory attribution of the same computation to different network components. Meloux et al. [2025] found **85 distinct valid circuits** for a single XOR task; Bricken et al. [2025] show sparse autoencoders learn different features per seed.

By Theorem 4, no circuit explanation can simultaneously be faithful, stable, and decisive. The resolution is to report the circuit equivalence class—features invariant across all valid decompositions form the "skeleton"; decomposition-dependent features are the "reversible edges." For AI safety: arguments that a specific circuit is THE mechanism of a capability must account for alternative circuits with potentially different safety properties.

Table 1: The three escape routes from the explanation trilemma.

Drop	Lose	Gain	Tools
Faithful	May contradict	Stable + decisive	Fixed rules
Stable	Changes on retrain	Faithful + decisive	SHAP, single circui
Decisive	Ties, uncertainty	Faithful + stable	Dash, CPDAG

3.4 Neural Degeneracy (Neuroscience)

The 6-tuple: $\mathbf{\Theta}$ = neural circuit parameterizations (ionic conductances, synaptic weights); Y = motor pattern output; $\mathcal{H}$ = mechanistic accounts of which ion channels drive the rhythm; $\perp$ = contradictory primary-channel attributions. Marder and Goaillard Marder and Goaillard [2006] documented that crustacean stomatogastric ganglia with identical motor patterns arise from qualitatively different conductance profiles—*degeneracy* Edelman and Gally [2001]—which is precisely the Rashomon property.

By Theorem 4, no mechanistic account can simultaneously be faithful to the specific circuit, stable across degenerate circuits, and decisive about the primary ionic mechanism. Biological and artificial neural circuits face the same structural impossibility.

The remaining five instances—attention maps (60% argmax flip rate across equivalent DistilBERT models), counterfactual explanations (23.5% direction flip rate), concept probes (0.90 direction instability), model selection (80% best-model flip rate), and token citation explanations (34.5% citation flip rate)—are detailed with full experimental methodology in *SI Appendix*.

4 Resolution

4.1 Escape Routes

The impossibility is tight: one desideratum must be sacrificed. Table 1 summarizes the three escape routes.

4.2 G-Invariant Aggregation

Let G be a group acting on the configuration space such that $g \cdot \theta$ preserves observables: $\mathsf{observe}(g \cdot \theta) = \mathsf{observe}(\theta)$. An explanation map is *G-invariant* if $E(g \cdot \theta) = E(\theta)$ for all g, θ.

Proposition 8 (G-Invariance Implies Stability)**.** *If G acts transitively on observe-fibers and E is G-invariant, then E is stable.*

When G is finite and the explanation space admits averaging, the canonical G-invariant map is the *orbit average*:

$$\bar{E}(\theta) = \frac{1}{|G|} \sum_{g \in G} \mathsf{explain}(g \cdot \theta).$$

This map is stable (by G-invariance), faithful in expectation (the expected explanation matches the expected

native explanation over the orbit), and Pareto-optimal among stable maps. It sacrifices decisiveness: where the orbit disagrees, the average produces ties.

The connection to classical statistics is direct. The Hunt–Stein theorem Lehmann and Romano [2005] establishes that among all invariant decision procedures, the best invariant procedure is admissible. Our G-invariant resolution is the explanation-theoretic instantiation: among all stable explanation maps, the orbit average achieves the best possible faithfulness.

4.3 Instance-Specific Resolutions

The abstract resolution instantiates concretely across domains:

- **Feature attribution**: DASH (**D**iversified **A**ggregation of **SH**AP)—average SHAP values over an ensemble of retrained models. Consensus equity ($\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features) is proved in Lean 4.
- **Causal discovery**: report the CPDAG (completed partially directed acyclic graph) rather than a single DAG Chickering [2002]. Undirected edges mark the unresolvable queries.
- **Mechanistic interpretability**: report the circuit equivalence class—the set of all valid circuits—rather than a single decomposition. Invariant circuit features form the "skeleton."
- **Neural degeneracy**: characterize the *degenerate set* of ionic conductance profiles producing the target motor pattern, rather than committing to a single mechanistic account.
- **Medical diagnosis**: differential diagnosis *is* G-invariant resolution—reporting the set of conditions consistent with the symptoms rather than committing to a single diagnosis.

5 Empirical Validation

We validate the impossibility empirically across seven of the nine instances (Table 2). For each instance, we train multiple models that are functionally equivalent (matched on prediction accuracy) and measure explanation instability.

Three results deserve emphasis. First, the 60% attention argmax flip rate: despite 100% prediction agreement among 10 DistilBERT models on SST-2, the most-attended token changes for the majority of sentences. This is not noise—it is the Rashomon property in action. Second, the 0.90 concept direction instability: concept activation vectors extracted from 15 equivalent MLP classifiers are nearly orthogonal (mean $|\cos| = 0.10$), meaning the "concept direction" is almost entirely a function of the random seed, not the function the model computes. Third, the 0% instability for negative controls (instances where the Rashomon property does not hold, e.g., unique-solution linear regression), confirming that the theorem's boundary condition is empirically sharp. The GradCAM 78.8% prediction agreement reflects sensitivity of ImageNet-pretrained ResNet-18 to perturbation on CIFAR-10. Four of twenty models in the attribution experiment achieved 72–78% held-out accuracy; all twenty are included in the analysis. Token citations are extracted via attention rollout as a proxy for LLM self-explanation.

Table 2: Cross-instance empirical validation. All instability measures have 95% bootstrap CIs excluding zero. Prediction equivalence confirms functional identity.

Instance	Metric	Value	Pred. equiv.
Attribution	SHAP flip	33%[‡]	>99%
Attention	Argmax flip	60%	100%
Counterfactual	Dir. flip	23.5%	ΔAUC<0.03
Concept probe	$1-\lvert\cos\rvert$	0.90	97.2%
Model selection	Best flip	80%	ΔAUC<0.02
Saliency map (GradCAM)	Peak flip	9.6%	78.8%[†]
Token citation	Citation flip	34.5%	100%[§]

[†]Prediction agreement reflects sensitivity of ImageNet-pretrained ResNet-18 to perturbation on CIFAR-10.
[‡]Four of twenty models achieved 72–78% held-out accuracy; all twenty are included in the analysis.
[§]Token citation extracted via attention rollout as a proxy.

Full experimental details—datasets, model architectures, perturbation protocols, confidence intervals, and negative controls—are in *SI Appendix*.

5.1 Information-Theoretic Perspective

Proposition 9 (Information-Theoretic Cost of Stability)**.** (Argued; paper proof.) *By the data processing inequality applied to* $\mathsf{explain}(\mathbf{\Theta}) \to \mathbf{\Theta} \to Y \to E$*, any stable map satisfies* $I(E; \mathsf{explain}(\mathbf{\Theta})) \leq I(Y; \mathsf{explain}(\mathbf{\Theta}))$*. When* $H(\mathsf{explain}(\mathbf{\Theta}) \mid Y) > 0$*, this bound is strictly less than* $H(\mathsf{explain}(\mathbf{\Theta}))$*, so any stable* E *forfeits at least* $H(\mathsf{explain}(\mathbf{\Theta}) \mid Y)$ *bits of the native explanation. This complements Theorem 4: both identify the Rashomon property as root cause, but the information-theoretic bound additionally quantifies the minimum information loss.*

6 Discussion

6.1 Implications Beyond Machine Learning

Tier 1: Formal instances. The theorem applies directly to any domain where the ExplanationSystem structure is instantiated. Beyond the nine ML instances, the database view update problem Bancilhon and Spyratos [1981] and Rashomon sets in statistics Fisher et al. [2019]

share the exact mathematical structure: multiple configurations (base tables, near-optimal models) map to the same observable (view, predictive performance) yet differ in their explanations (update translation, variable importance rankings).

Tier 2: Substantive applications requiring formalization. The degeneracy principle in neuroscience Marder and Goaillard [2006], Edelman and Gally [2001], partial identification in econometrics Manski [2003], and differential diagnosis in medicine exhibit the same structure: multiple configurations produce identical observables. Formalizing each as an ExplanationSystem and verifying the Rashomon property would yield the impossibility as a corollary. We leave these formalizations to domain specialists, but note that the neuroscience instance is treated in Section 3.4 and the medical diagnosis resolution (differential diagnosis as G-invariant aggregation) is described in Section 4.

Tier 3: Suggestive connections (discussion only). The Quine–Duhem underdetermination thesis Quine [1951], Duhem [1954] is the informal precursor to our result. Theorem 4 can be read as a formalization of a core consequence of underdetermination: no method of choosing between empirically equivalent theories can simultaneously be faithful to a specific theory, stable across equivalent theories, and decisive about theoretical commitments. The G-invariant resolution formalizes what structural realists Worrall [1989] have argued informally: the stable content of a theory is its structure, not its interpretation. We note, however, that the philosophical setting involves normative and ontological dimensions that our descriptive framework does not capture. Okasha Okasha [2011] connected theory choice to Arrow's impossibility theorem Arrow [1951]; our result provides the complementary connection for underdetermination of explanation.

6.2 Regulatory Analysis

OCC SR 11-7 Office of the Comptroller of the Currency [2011] maps onto the trilemma: Conceptual Soundness demands *faithfulness*, Outcomes Analysis across model versions demands *stability*, and Effective Challenge demands *decisiveness*. These three pillars cannot be simultaneously satisfied for any model with the Rashomon property—generic for models trained with bagging, boosting, or SGD. Compliant documentation should state which property is prioritized, use ensemble explanations (Dash) for the stable+faithful tradeoff, and report Rashomon set size and explanation variance. The same analysis applies to EU AI Act Article 13 EU Parliament [2024] and NIST AI RMF MAP 1.5 National Institute of Standards and Technology [2023]: each regulatory framework implicitly assumes all three properties are jointly achievable; Theorem 4 shows they are not. Actionable guidance for common practitioner goals is given in *SI Appendix*, Table S2.

6.3 AI Safety

The mechanistic interpretability instance (Section 3.3) has direct implications for AI safety. If a safety argument depends on a specific circuit being the mechanism of a dangerous capability, the impossibility theorem requires that the argument account for alternative circuits—functionally equivalent decompositions that may have different safety-relevant properties. The 85 valid circuits found by Meloux et al. [2025] for a simple XOR task suggest that real-world safety-critical capabilities will admit far more. The constructive response is to build safety arguments on circuit-invariant properties—features that hold across the entire equivalence class—rather than on any single decomposition.

6.4 Closing

No explanation of an underspecified system can simultaneously be faithful, stable, and decisive. The impossibility does not counsel despair but precision: state which property is sacrificed, aggregate optimally, and report uncertainty. The Rashomon property is the exact boundary. When it holds, the trilemma is inescapable. When it does not, all three properties are jointly achievable. The task for the practitioner is to diagnose which regime obtains and to communicate the tradeoff honestly.

Materials and Methods

The Lean 4 formalization (74 files, 349 theorems, 72 axioms, 0 `sorry`) is available at [repository URL]. Experimental code and data are in *SI Appendix*. The core impossibility theorem (`explanation_impossibility`) depends on zero axioms beyond the Rashomon property. Full axiom inventory and theorem list are in *SI Appendix*, Table S1.

Acknowledgments

[To be added.]

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